

An Efficient IDS System for IoT Attacks

1st Dr. Josy Elsa Varghese

Dept. of CSE (AIML & Cyber Security)
M. S. Ramaiah Institute of Technology
Bangalore, India

2nd Vignesh K

Dept. of CSE (Cyber Security)
M. S. Ramaiah Institute of Technology
Bangalore, India

3rd V Sravan Reddy

Dept. of CSE (Cyber Security)
M. S. Ramaiah Institute of Technology
Bangalore, India

4th Jaliha Abhay

Dept. of CSE (Cyber Security)
M. S. Ramaiah Institute of Technology
Bangalore, India

Abstract—The rapid growth of Internet of Things (IoT) devices and edge-based communication systems has created major security challenges in modern networks. Most traditional security systems mainly depend on static rules and classical cryptographic algorithms which are becoming insufficient against advanced cyberattacks and future quantum computing threats. This project presents an AI-powered network intrusion detection system integrated with lightweight post-quantum secure communication for IoT edge environments. The developed framework continuously captures live network packets using Scapy and converts them into flow-based statistical features for machine learning analysis. A calibrated Extra Trees based machine learning model was trained using balanced datasets generated with SMOTE to improve intrusion detection performance across multiple attack categories. The system is capable of detecting attacks such as DoS, DDoS, scanning attacks, brute-force attempts, exploit-based attacks, and other anomalous traffic patterns. To improve communication security, a lightweight post-quantum cryptography layer was integrated for secure key exchange while AES-256-GCM encryption was used for protecting payload communication. The complete system was evaluated using real-time attack traffic generated from virtual machines and edge devices including Raspberry Pi. Experimental results show that the proposed system achieved high detection accuracy with moderate runtime overhead while maintaining lightweight resource usage suitable for IoT and edge deployment.

Index Terms—Network Intrusion Detection System, Post-Quantum Cryptography, IoT Security, Machine Learning, Edge Computing, Raspberry Pi, AES-GCM, Real-Time Monitoring

I. INTRODUCTION

The increasing dependency on IoT devices and edge-based communication systems has created a large attack surface for modern cyber threats. IoT devices are widely used in health-care systems, smart homes, industrial automation, transportation systems, and smart city applications. Since these devices continuously exchange data through networks, attackers can target them using different methods such as port scanning, denial-of-service attacks, brute-force login attempts, packet injection, and exploit-based traffic.

Traditional network security solutions mainly rely on static signatures and predefined firewall rules. Although these systems are useful for detecting known attacks, they are not always effective against newly generated attack patterns or unknown traffic behaviour. In many practical cases, attackers

slightly modify the attack flow so that the traffic does not exactly match fixed signatures. This makes network monitoring more difficult, especially in IoT environments where devices are limited in processing power and memory.

A Network-Based Intrusion Detection System (NIDS) improves security by monitoring packet flow across the network instead of depending only on one host device. A NIDS checks traffic behaviour, packet count, flow duration, source and destination details, protocol type, and timing values. With machine learning, the IDS can learn patterns from normal and attack traffic and classify suspicious activity more effectively.

Another important issue is the growth of quantum computing. Classical cryptographic systems such as RSA and ECC may become weak in the future when large-scale quantum computers become practical. This creates the need for post-quantum cryptography, which is designed to resist both classical and quantum-based attacks. In this project, a lightweight post-quantum key exchange concept is used along with AES-256-GCM encryption to improve communication security.

The main aim of this project is to build a practical network-based IDS that can detect attacks in real time while also supporting secure communication. The system was developed using Python, Scapy, Scikit-learn, Streamlit, Raspberry Pi, and Kali Linux testing tools. The major contributions of this project are:

- A real-time machine learning-based network IDS was developed.
- Live packets were captured and converted into flow-based features.
- SMOTE was used to balance attack classes during model training.
- Threshold tuning was applied to improve detection stability.
- A lightweight PQC-inspired key exchange layer was added.
- AES-256-GCM encryption was used for secure payload protection.
- A Streamlit dashboard was developed for real-time monitoring.
- Raspberry Pi was used as an edge gateway in the deployment setup.

II. RELATED WORK

Intrusion detection has been an important area in network security for many years. Early IDS systems mainly used rule-based and signature-based detection methods. These systems compare traffic with known attack patterns and generate alerts when a match is found. This approach is simple and fast, but it cannot detect new attacks that do not already exist in the rule database.

Machine learning-based IDS systems solve this problem by learning traffic behaviour from datasets. Algorithms such as Decision Tree, Random Forest, Support Vector Machine, Extra Trees, and Neural Networks have been used for classifying network traffic. These models can learn patterns from packet counts, byte counts, flow duration, and inter-arrival timings. ML-based IDS systems are more suitable for detecting unknown or modified attack patterns compared to traditional rule-based IDS systems.

Several IoT security studies have also used deep learning methods for anomaly detection. However, many deep learning models are computationally heavy and may not run efficiently on small edge devices. In this project, an Extra Trees based model was selected because it gives good classification performance while keeping runtime overhead manageable. Post-quantum cryptography has also become important due to future quantum threats. Most existing projects focus either on IDS detection or cryptography separately. This project combines both network-based intrusion detection and post-quantum secure communication into one edge-oriented framework.

III. SYSTEM ARCHITECTURE

The proposed system follows a layered architecture where each layer performs a specific task. The architecture includes traffic collection, secure communication, edge processing, attack detection, adaptive thresholding, and dashboard visualization.

A. Overall Architecture

The architecture diagram of the proposed system is shown in Fig. 1. The system starts from IoT devices, sensors, and network datasets. Raw packets are captured and forwarded to the edge processing layer. The network security layer performs post-quantum key exchange and AES-GCM encryption.

B. Perception Layer

The perception layer is responsible for collecting raw network traffic. In the proposed system, this traffic can come from IoT devices, Raspberry Pi, virtual machines, and dataset-based traffic sources such as ToN-IoT and BoT-IoT. Packet-level details such as source IP, destination IP, protocol type, source port, destination port, packet size, and timing information were extracted.

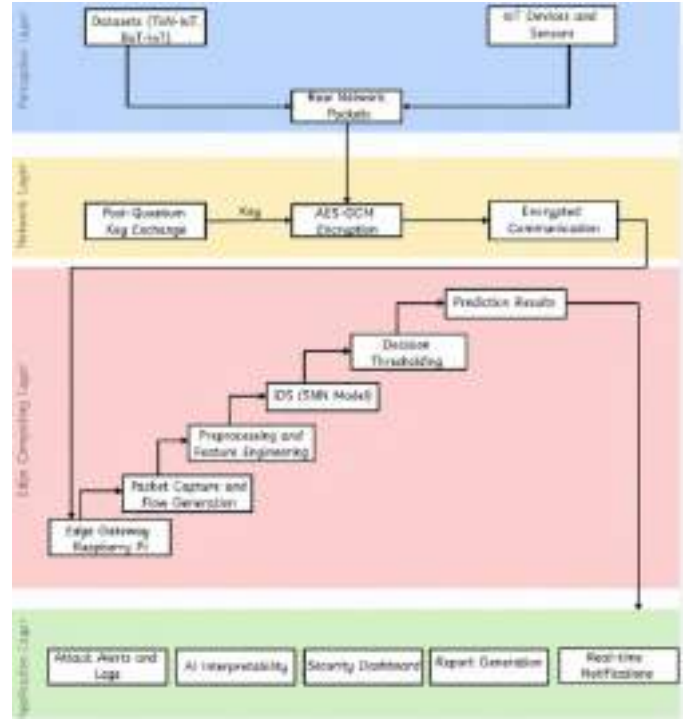


Fig. 1. Proposed AI-Powered Network IDS Architecture

C. Network Security Layer

The network security layer provides communication protection. A lightweight post-quantum style key exchange layer is used to simulate quantum-resistant key establishment. Once the session key is generated, AES-256-GCM encryption is used for protecting payload data.

D. Edge Computing Layer

The edge computing layer is the main processing layer. It receives packets, creates flows, extracts features, applies preprocessing, and performs attack classification using the trained IDS model. The Raspberry Pi acts as an edge gateway in the system.

E. Application Layer

The application layer provides the user interface and monitoring support. A Streamlit dashboard was created to display runtime information such as packet count, active flows, alerts, attack type, confidence, backend status, and PQC status.

IV. THREAT MODEL AND COUNTERMEASURES

The project mainly focuses on common network-based attacks that can affect IoT and edge environments. Table I shows the major threats considered in the proposed system and the countermeasures used.

V. DATASET AND PREPROCESSING

The IDS model was trained using a merged intrusion detection dataset containing approximately 299,998 records and seven traffic classes: benign, ddos, dos, scan, exploit, bruteforce, and other attack.

TABLE I
THREAT MODELS AND COUNTERMEASURES

Threat	Impact	Countermeasure
DoS	Slowdown or packet loss	Flow rate monitoring and ML detection
DDoS	Network congestion	Source tracking and DDoS thresholds
Scan	Finds weak services	Port variation and scan detection
Brute Force	Access risk	Attempt pattern detection
Exploit	System compromise	Feature-based ML detection
Quantum	Loss of confidentiality	PQC-style key exchange and AES-GCM
False Alerts	Unwanted alerts	Threshold tuning and confidence logic

A. Dataset Distribution

The original dataset was imbalanced, which can bias the model toward majority classes. To reduce the effect of class imbalance, SMOTE was applied to create synthetic samples for minority classes.

TABLE II
DATASET DISTRIBUTION BEFORE SMOTE

Class	Count	Percentage
Benign	181163	60.39%
Exploit	30285	10.10%
Scan	25392	8.46%
DDoS	22702	7.57%
DoS	20960	6.99%
Other Attack	17394	5.80%
Bruteforce	2102	0.70%

TABLE III
DATASET DISTRIBUTION AFTER SMOTE

Class	Count
Benign	145574
DDoS	24263
DoS	24263
Scan	24263
Exploit	24263
Other Attack	24263
Bruteforce	24263

B. Preprocessing Steps

The preprocessing stage included missing value handling, feature selection, label encoding, scaling, and dataset splitting. StandardScaler was used to normalize numerical features.

VI. FEATURE ENGINEERING

Feature engineering plays an important role because the IDS does not directly classify raw packets. Instead, packets are grouped into flows, and numerical features are extracted to represent the behaviour of network traffic. Important features include packet counts, byte counts, inter-arrival timings, and flow duration.

Average forward segment size and flow duration contributed strongly toward attack detection. Features related to backward

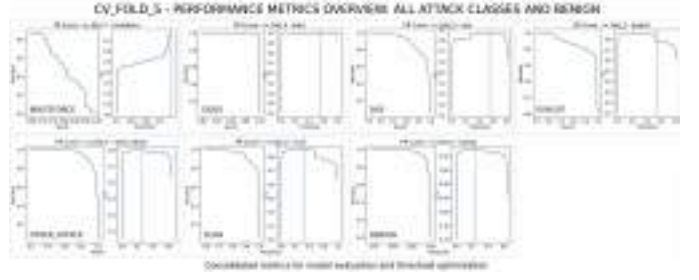


Fig. 2. Consolidated Performance Metrics Overview for IDS Classification

inter-arrival time and packet counts also showed significant importance in identifying abnormal traffic behaviour.

VII. MACHINE LEARNING MODEL

The intrusion detection model was developed using a lightweight ensemble-based model suitable for real-time edge deployment.

A. Extra Trees Classifier

The Extra Trees ensemble classifier works using multiple randomized decision trees. It provides high classification accuracy, fast runtime inference, and works effectively for multiclass traffic classification.

B. Probability Calibration and Threshold Tuning

Probability calibration was applied using CalibratedClassifierCV with sigmoid scaling. Instead of directly using maximum probability prediction, the system compares probabilities with per-class thresholds to reduce false positives.

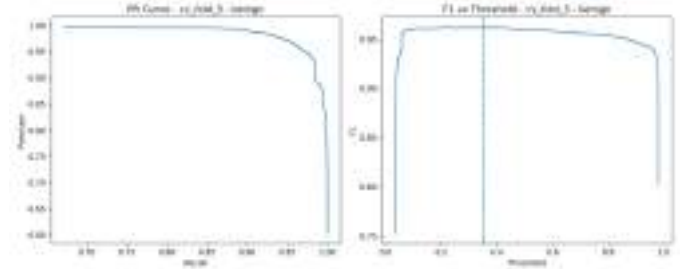


Fig. 3. Precision-Recall Curve for Benign Traffic

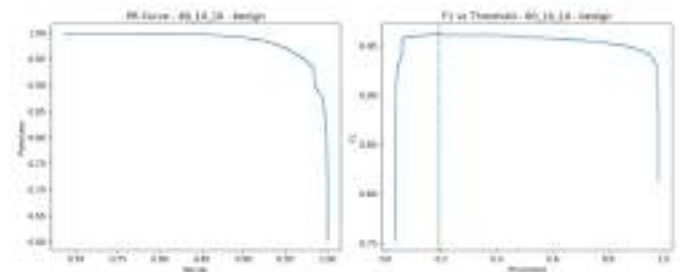


Fig. 4. F1 Score vs Threshold Analysis

The PR curve shows that the trained model maintains high precision and recall values. The threshold graph was used to determine optimal runtime confidence values. Cross-validation experiments showed stable classification behaviour across multiple dataset splits.

VIII. REAL-TIME DETECTION WORKFLOW

The runtime system continuously monitors live network traffic using Scapy. Packets are grouped into host-aware flows and converted into statistical feature vectors. The IDS supports real-time detection of DoS, DDoS, Port scanning, Brute-force attempts, and Exploit traffic.

IX. POST-QUANTUM SECURE COMMUNICATION

To address future quantum computer threats, this project integrates a PQC-inspired secure key exchange mechanism.

A. PQC Key Exchange & AES-256-GCM Encryption

The PQC layer simulates lattice-based key exchange concepts inspired by Kyber-style algorithms. After secure key exchange, AES-256-GCM encryption is used for data confidentiality, authentication support, and fast encryption speed.

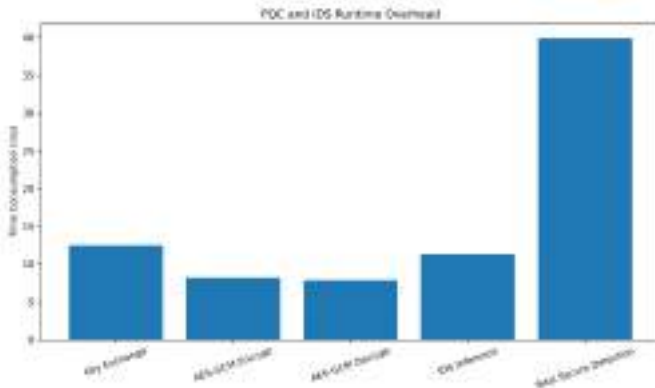


Fig. 5. PQC and IDS Runtime Overhead

The graph shows that the major runtime cost comes from secure detection and inference operations, while AES-GCM encryption and decryption maintain relatively low overhead.

X. EXPERIMENTAL SETUP

The project was tested using both Windows and Raspberry Pi environments, with Kali Linux virtual machines generating attack traffic.

TABLE IV
HARDWARE CONFIGURATION

Component	Specification
Laptop	Intel i5, 16GB RAM
Raspberry Pi	Raspberry Pi 4 Model B
Operating System	Windows 11 / Raspberry Pi OS
Virtual Machine	Kali Linux
Programming Language	Python 3.11

TABLE V
SOFTWARE TOOLS USED

Tool	Purpose
Python	Main implementation
Scapy	Packet capture
Scikit-learn	Machine learning
Pandas	Data preprocessing
Streamlit	Dashboard UI
Plotly	Visualization
Kali Linux	Attack generation
Nmap	Scan testing

XI. RESULTS AND PERFORMANCE ANALYSIS

The proposed IDS was evaluated using live traffic and dataset-based testing.

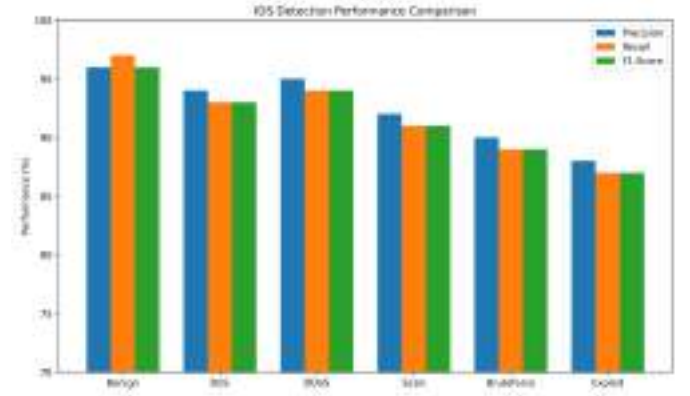


Fig. 6. IDS Detection Performance Comparison

The model achieved strong performance across multiple attack categories. Benign traffic achieved the highest F1-score because of larger dataset representation.

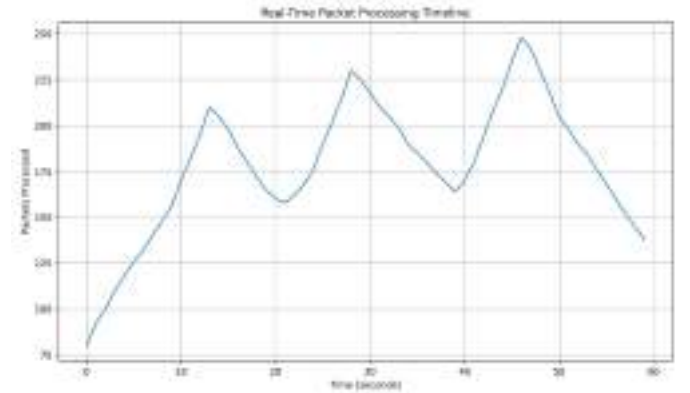


Fig. 7. Real-Time Packet Processing Timeline

The graph shows stable packet processing capability over time. Temporary spikes occurred during simulated attack bursts, but the system maintained continuous monitoring.

The results show that the proposed system remains lightweight enough for edge deployment. CPU usage remained moderate while RAM usage stayed within practical limits.

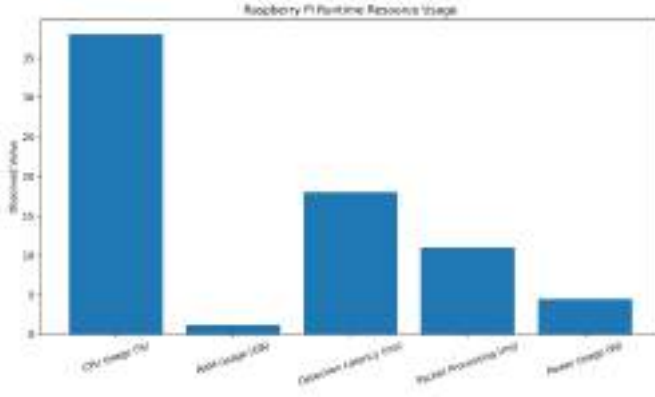


Fig. 8. Raspberry Pi Runtime Resource Usage

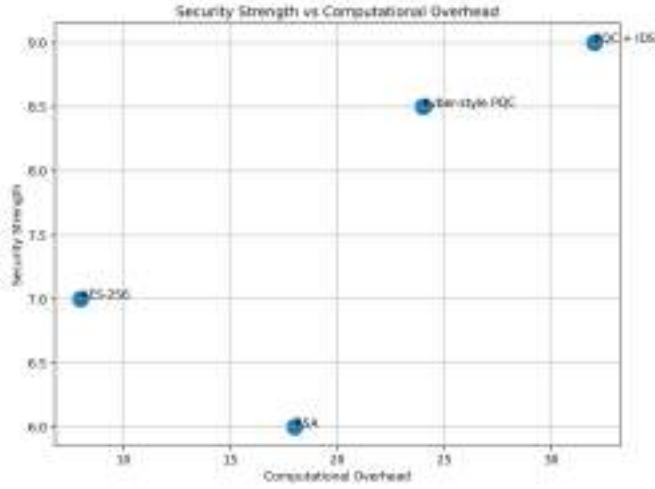


Fig. 9. Security Strength vs Computational Overhead

The integrated PQC and IDS framework provides stronger security compared to classical RSA-based communication, making it highly suitable for edge deployment.

XII. ADVANTAGES OF PROPOSED SYSTEM

The developed system provides several practical advantages: Real-time attack monitoring, lightweight edge deployment, flow-based anomaly detection, reduced false positive alerts, quantum-resistant communication support, dashboard-based monitoring, multi-attack detection capability, secure payload encryption, and adaptive threshold-based decisions.

XIII. LIMITATIONS

The PQC implementation is lightweight and partially simulated. Real-world enterprise traffic is more complex than laboratory testing. Encrypted traffic analysis remains difficult, and heavy DDoS attacks may require distributed edge deployment.

XIV. CONCLUSION

This project presented a real-time AI-powered Network Intrusion Detection System integrated with lightweight post-quantum secure communication for IoT edge environments.

The proposed framework combines machine learning-based attack detection, adaptive threshold logic, secure payload encryption, and edge deployment support into one unified architecture. Experimental results demonstrated strong precision, recall, and F1-score performance across multiple traffic classes. Overall, the project demonstrates that combining machine learning intrusion detection with lightweight post-quantum secure communication can improve IoT and edge network security without requiring heavy computational infrastructure.

XV. FUTURE WORK

Several improvements can be added in future versions: Integration of official NIST-standardized PQC algorithms, deep learning-based behavioural traffic analysis, federated learning across multiple edge devices, cloud-assisted distributed IDS deployment, and automated response mitigation.

REFERENCES

- [1] L. S. Morin and B. A. Princy, "Lightweight Cryptography and Intrusion Detection System for Edge Networks," IEEE, 2025.
- [2] A. Sharma and S. Rani, "Post-Quantum Cryptography for IoT-Consumer Electronics Devices Integrated with Deep Learning," IEEE, 2025.
- [3] J. E. Varghese and V. Rao, "A Survey on DDoS Attacks in IoT Cloud Platforms," IEEE, 2024.
- [4] G. P., J. E. Varghese, and S. G. Aarella, "A Survey on Anomaly Detection in IoT and Cloud Computing Security," IEEE, 2024.
- [5] S. Duraibi and A. M. Alashjaee, "Lightweight Post-Quantum Secure Communication Protocol for IoT Devices Using Code-Based Cryptography," IEEE, 2025.
- [6] A. Pasam, A. Sinha, and A. Mailewa, "Lightweight AI-Based Intrusion Detection Models for IoT Devices: A Comparative Review," IEEE, 2025.
- [7] N. O. Gavrilidis et al., "Identity-Based Cryptography on IoT Devices: Balancing Security and Efficiency for Modern Use Cases," IEEE, 2024.
- [8] Z. Alomari, Z. Li, and A. Makanju, "Lightweight Machine Learning-Based Intrusion Detection System for IoT Environments," IEEE, 2024.
- [9] M. S. Khaing, Z. M. Aye, and T. Tun, "Machine Learning-Based Performance Analysis for IoT Attacks in Intrusion Detection Systems," IEEE, 2023.
- [10] M. A. Mahdi, "Secure and Efficient IoT Networks: An AI and Machine Learning-Based Intrusion Detection System," IEEE, 2024.
- [11] D. Aggarwal, A. B. Saxena, and D. Sharma, "Mitigating Cybersecurity Risks in IoT: A Layered Approach to Threat Detection and Prevention," IEEE, 2025.
- [12] A. B. Gadewar, R. V. Patil, S. A. Mahajan, and L. V. Patil, "Automated Network Vulnerability Detection in IoT: Lightweight IDS Approaches for Enhanced Security," IEEE, 2024.
- [13] T. Azzabi, M. A. Kacem, M. H. Jeridi, and T. Ezzedine, "Enhancing IoT Network Security through Intelligent Device Identification Using Machine Learning," IEEE, 2025.
- [14] S. Yoon, B. Kim, K. Kim, and Y. Kang, "Enhancing IoT Security with PUF-Based Authentication Scheme," IEEE, 2022.
- [15] K. Abinaya, T. Lohith, and S. Jayanth Kumar, "Enhancing Network Security with Intrusion Detection Systems in IoT Devices," IEEE, 2025.